Nepal Federal Election 2022 – Data Analysis Project Report

This report documents the end-to-end data preparation and exploratory analysis process performed on the Nepal Federal Election 2022 dataset. The project demonstrates practical data analyst skills including data acquisition, cleaning, transformation, exploratory analysis, and preparation for SQL and BI tools.

1. Project Objective

The goal of this project was to analyze Nepal's Federal Election 2022 results using a data-driven workflow. The project focuses on transforming raw election data into an analysis-ready dataset and generating meaningful insights about party performance, constituency competition, and electoral trends.

2. Data Source

The dataset was collected from the Election Commission of Nepal. The data contains candidate-level election results for the Federal Parliament (First-Past-The-Post system).

3. Project Folder Structure

The project used a professional analytics structure:

Data/raw – original datasets
Data/clean – processed datasets
notebooks – Python notebooks used for analysis

4. Data Cleaning Steps

The raw dataset required several cleaning steps:
• Standardized column names
• Removed translated columns
• Trimmed whitespace and handled missing values
• Converted vote counts to numeric values
• Removed duplicates
• Added election year column

5. Constituency Reconstruction

The dataset stored only constituency numbers within districts. A new column combining district and constituency number was created to form a unique constituency identifier.

6. Analytical Outputs

The analysis generated multiple insights including:

- Total vote count
- Party vote share
- Seat share vs vote share comparison
- Closest electoral races
- Province-wise seat distribution
- Gender participation
- Independent candidate performance

7. Visualization

Charts were created using Matplotlib to visualize party performance, seat distribution, and election competitiveness.

8. Skills Demonstrated

This project demonstrates skills in:

- Data cleaning
- Feature engineering
- Exploratory analysis
- Data visualization
- Preparing datasets for SQL databases

9. Conclusion

This project demonstrates a complete data analysis workflow using real election data and prepares the dataset for deeper analysis using SQL and BI tools.